6-3

1. For which of the following integrals is $\sum_{k=1}^{n} \ln\left(2+\frac{5}{n}k\right)\frac{5}{n}$ the right Riemann sum approximation with n subintervals of equal length?

(A) $\int_2^7 \ln(x)\,dx$

(B) $\int_2^7 \ln(2+x)\,dx$

(C) $\int_0^5 \ln(2+5x)\,dx$

(D) $\int_2^7 \ln(2+5x)\,dx$

Answer A

Correct. The sum can be interpreted as a right Riemann sum in the form $\sum_{k=1}^{n} f(2+k\Delta x)\Delta x$, where $f(x)=\ln(x)$ and $\Delta x=\frac{5}{n}$. The value of Δx corresponds to an interval of length 5. The sum starts with the right endpoint $2+\Delta x$ and ends with the right endpoint $2+n\Delta x=2+5=7$, so the Riemann sum is over the interval $[2,7]$. This would be the right Riemann sum approximation for the definite integral $\int_2^7 f(x)\,dx=\int_2^7 \ln(x)\,dx$.

6-3

2.

x	$f(x)$
-5	-9
0	1
2	5

The table above gives values of a continuous function f at selected values of x.

Based on the information in the table, which of the following statements must be true?

(A) f has at most one zero.

(B) f has a relative maximum at $x = 2$.

(C) There exists a value c, where $-5 < c < 2$, such that $f(c) = 4$.

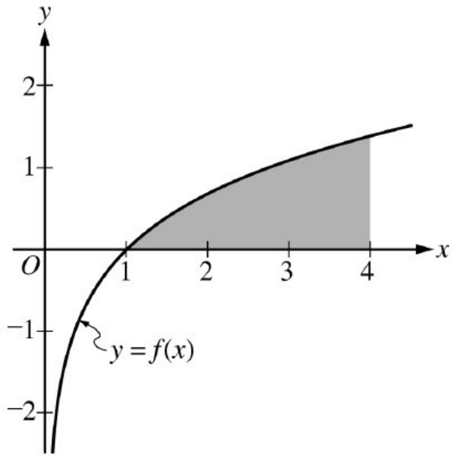
(D) There exists a value c, where $-5 < c < 2$, such that $f'(c) = 2$.

Answer C

Correct. Since f is continuous on the closed interval $[-5, 2]$ and $f(-5) < 4 < f(2)$, then by the Intermediate Value Theorem there must be a value c in the open interval $(-5, 2)$ such that $f(c) = 4$.

6-3

3.



The function f is given by $f(x) = \ln x$. The graph of f is shown above. Which of the following limits is equal to the area of the shaded region?

(A) $\lim_{n\to\infty} \sum_{k=1}^{n} \left(1 + \ln\left(\frac{3k}{n}\right)\right)\frac{3}{n}$

(B) $\lim_{n\to\infty} \sum_{k=1}^{n} \ln\left(1 + \frac{3k}{n}\right)\frac{3}{n}$

(C) $\lim_{n\to\infty} \sum_{k=1}^{n} \ln\left(\frac{4}{n}\right)\left(1 + \frac{4k}{n}\right)$

(D) $\lim_{n\to\infty} \sum_{k=1}^{n} \ln\left(1 + \frac{4k}{n}\right)\frac{4}{n}$

Answer B

This option is correct. The area of the shaded region is equal to $\int_1^4 \ln x \, dx$. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{4-1}{n} = \frac{3}{n}$. The height of each rectangle is determined by the functions ln x evaluated at the right endpoints of the n intervals. These endpoints are the x values at $1 + k\Delta x$, for k from 1 to n, since the integral starts at 1. The Riemann sum can be written as $\sum_{k=1}^{n} \ln(1 + k\Delta x)\Delta x$ where $\Delta x = \frac{3}{n}$, and its limit as $n \to \infty$ is $\int_1^4 \ln x \, dx$.

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4. A solid has a rectangular base that lies in the first quadrant and is bounded by the x- and y-axes and the lines $x = 2$ and $y = 1$. The height of the solid above the point (x, y) is $1 + 3x$. Which of the following is a Riemann sum approximation for the volume of the solid?

(A) $\sum_{i=1}^{n} \frac{1}{n}\left(1+\frac{3i}{n}\right)$

(B) $2\sum_{i=1}^{n} \frac{1}{n}\left(1+\frac{3i}{n}\right)$

(C) $2\sum_{i=1}^{n} \frac{i}{n}\left(1+\frac{3i}{n}\right)$

(D) $\sum_{i=1}^{n} \frac{2}{n}\left(1+\frac{6i}{n}\right)$ ✓

(E) $\sum_{i=1}^{n} \frac{2i}{n}\left(1+\frac{6i}{n}\right)$

5. For a certain continuous function f, the right Riemann sum approximation of $\int_0^2 f(x)\,dx$ with n subintervals of equal length is $\frac{2(n+1)(3n+2)}{n^2}$ for all n. What is the value of $\int_0^2 f(x)\,dx$?

(A) 2

(B) 6 ✓

(C) 12

(D) 20

6-3

6. Which of the following limits is equal to $\int_2^5 x^2 dx$

(A) $\lim_{n\to\infty}\sum_{k=1}^{n}\left(2+\frac{k}{n}\right)^2\frac{1}{n}$

(B) $\lim_{n\to\infty}\sum_{k=1}^{n}\left(2+\frac{k}{n}\right)^2\frac{3}{n}$

(C) $\lim_{n\to\infty}\sum_{k=1}^{n}\left(2+\frac{3k}{n}\right)^2\frac{1}{n}$

(D) $\lim_{n\to\infty}\sum_{k=1}^{n}\left(2+\frac{3k}{n}\right)^2\frac{3}{n}$ ✓

Answer D

This option is correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x=\frac{5-2}{n}=\frac{3}{n}$. The height of each rectangle is determined by the function x^2 evaluated at the right endpoints of the n intervals. These endpoints are the x-values at $2+k\Delta x$, for k from 1 to n, since the integral starts at 2. The Riemann sum can be written as $\sum_{k=1}^{n}(2+k\Delta x)^2\Delta x$ where $\Delta x=\frac{3}{n}$, and its limit as $n\to\infty$ is $\int_2^5 x^2 dx$.

7. The closed interval $[a,b]$ is partitioned into n equal sub intervals, each of width Δx , by the numbers $x_0, x_1, \cdots, x_n$ where

$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$. What is $\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{x_i}\Delta x$?

(A) $\frac{2}{3}\left(b^{\frac{3}{2}}-a^{\frac{3}{2}}\right)$ ✓

(B) $b^{\frac{3}{2}}-a^{\frac{3}{2}}$

(C) $\frac{3}{2}\left(b^{\frac{3}{2}}-a^{\frac{3}{2}}\right)$

(D) $b^{\frac{1}{2}}-a^{\frac{1}{2}}$

(E) $2\left(b^{\frac{1}{2}}-a^{\frac{1}{2}}\right)$

6-3

8. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	4	5	8	10	14
$h'(t)$ (feet per hour)	0.3	0.6	0.7	1.4	2.3	7.7

The height of water in a storage tank during a 15-hour period is given by a twice-differentiable function h, where $h(t)$ is measured in feet and t is measured in hours since midnight for $0 \le t \le 15$. The graph of h is concave up on the interval $0 \le t \le 15$. Selected values of the derivative of h, $h'(t)$, are given in the table above. At time $t = 2$, the height of the water is 7 feet.

(a) Use the tangent line approximation for h at time $t = 2$ to estimate $h(2.8)$, the height of the water at time $t = 2.8$. Is the approximation an overestimate or an underestimate for $h(2.8)$? Give a reason for your answer.

(b) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_2^{14} h'(t)\,dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or underestimate for $\int_2^{14} h'(t)\,dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^{n} h'\left(\frac{15}{n}(k-1)\right)\frac{15}{n}$ is a left Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 2.8$.

Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-3

0	1	2

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

$h\ (2.8) \approx h\ (2) + h'\ (2)\ (2.8 - 2) = 7 + 0.3\ (0.8) = 7.24$ feet

Since the graph of h is concave up on $2 \leq t \leq 2.8$, the tangent line lies under the graph of h on this interval. Therefore, the approximation is an underestimate for $h\ (2.8)$.

Part B

To earn the first point, the response must present the form of a right Riemann sum.

To earn the second point, values for h' must be substituted into the sum. Therefore, $2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ right Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned}\int_2^{14} h'\ (t)\ dt &\approx (4-2)h'\ (4) + (5-4)h'\ (5) + (8-5)h'\ (8) + (10-8)h'\ (10) + (14-10)h'\ (14)\\ &= 2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7 = 41.5 \text{ feet}\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-3

0	1

The student response accurately includes the criteria below.

- [ ] overestimate with reason that references the behavior of h'

Solution:

Since the graph of h is concave up on $0 \leq t \leq 15$, h' is increasing on $2 \leq t \leq 14$. Therefore, the right Riemann sum approximation is an overestimate for $\int_2^{14} h'(t)\ dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 15. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $\Delta t = \frac{15}{n}$
- [ ] limits of integration
- [ ] integrand

Solution:

For this Riemann sum, $\Delta t = \frac{15}{n}$.

Thus, the integral is over an interval of length This interval can be taken to be $0 \leq t \leq 15$.

The left endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{15}{n}(k-1)$ for k from 1 to n.

Therefore, $\lim_{n \to \infty} \left(\sum_{k=1}^{n} h'\left(\frac{15}{n}(k-1)\right) \frac{15}{n} \right) = \int_0^{15} h'(t)\ dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+15} h'(t-a)\ dt$.

6-3

9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$R(t)$ (gallons per hour)	16	12	7	3	0

The rate at which oil leaks from a drum is modeled by the twice-differentiable function R, where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \le t \le 2$. Values of $R(t)$ are given in the table above for selected values of t.

(a) Use the data in the table to find an approximation for $R'\left(\frac{3}{4}\right)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_0^2 R(t)\,dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^1 R'(t)\,dt$.

(d) The sum $\sum_{k=1}^{n} R\left(\frac{1}{2}+\frac{k}{n}\right)\frac{1}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

6-3

- ☐ approximation
- ☐ units

Solution:

$$R'\left(\tfrac{3}{4}\right) \approx \frac{R(1)-R(\frac{1}{2})}{1-\frac{1}{2}} = \frac{7-12}{\frac{1}{2}} = -10$$

gallons per hour per hour

Part B

To earn the first point, the response must present the form of a trapezoidal sum. Therefore, $\frac{1}{2}\left(\frac{1}{2}(16+12)+\frac{1}{2}(12+7)+\frac{1}{2}(7+3)+\frac{1}{2}(3+0)\right)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ units

Solution:

$$\int_0^1 R(t)\,dt \approx \tfrac{1}{2}\left(\tfrac{1}{2}(R(0)+R(\tfrac{1}{2}))+\tfrac{1}{2}(R(\tfrac{1}{2})+R(1))+\tfrac{1}{2}(R(1)+R(\tfrac{3}{2}))+\tfrac{1}{2}(R(\tfrac{3}{2})+R(2))\right)$$
$$= \tfrac{1}{2}\left(\tfrac{1}{2}(16+12)+\tfrac{1}{2}(12+7)+\tfrac{1}{2}(7+3)+\tfrac{1}{2}(3+0)\right)$$
$$= \tfrac{1}{4}(28+19+10+3) = 15 \text{ gallons}$$

Part C

The response should include supporting work for the numerical answer of -9.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

6-3

The student response accurately includes a correct value.

Solution:

$$\int_0^1 R'(t)\,dt = R(1) - R(0) = 7 - 16 = -9$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 1. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $\Delta t = \frac{1}{n}$
- [ ] limits of integration
- [ ] integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{n}$. Thus, the integral is over an interval of length 1. This interval can be taken to be $\left[\frac{1}{2}, \frac{3}{2}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{2} + \frac{k}{n}$ for k from 1 to n.

Therefore, $\lim_{n\to\infty}\left(\sum_{k=1}^{n} R\left(\frac{1}{2}+\frac{k}{n}\right)\cdot\frac{1}{n}\right) = \int_{\frac{1}{2}}^{\frac{3}{2}} R(t)\,dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+1} R\left(\frac{1}{2} - a + t\right)\,dt.$

6-3

10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{3}$	$\frac{2}{3}$	1
$R(t)$ (gallons per hour)	11	8	5	0

The rate at which water leaks from a container is modeled by the twice-differentiable function R, where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 1$. Values of $R(t)$ are given in the table above for selected values of t.

(a) Use the data in the table to find an approximation for $R'\left(\frac{1}{2}\right)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate $\int_0^1 R(t)\,dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^{\frac{1}{3}} R'(t)\,dt$.

(d) The sum $\sum_{k=1}^{n} R\left(\frac{1}{4}+\frac{k}{2n}\right)\frac{1}{2n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

6-3

- [ ] approximation
- [ ] units

Solution:

$$R'\left(\tfrac{1}{2}\right) \approx \frac{R\left(\frac{2}{3}\right)-R\left(\frac{1}{3}\right)}{\frac{2}{3}-\frac{1}{3}} = \frac{5-8}{\frac{1}{3}} = -9 \text{ gallons per hour per hour}$$

Part B

To earn the first point, the response must present the form of a left Riemann sum. Therefore, $\frac{1}{3}(11+8+5)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] left Riemann sum
- [ ] approximation
- [ ] units

Solution:

$$\int_0^1 R(t)\,dt \approx \tfrac{1}{3}\left(R(0)+R\left(\tfrac{1}{3}\right)+R\left(\tfrac{2}{3}\right)\right)$$
$$= \tfrac{1}{3}(11+8+5) = 8 \text{ gallons}$$

Part C

The response should include supporting work for the numerical answer of -3.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes a correct value.

Solution:

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$$\int_0^{\frac{1}{3}} R'(t)\,dt = R\left(\tfrac{1}{3}\right) - R(0) = 8 - 11 = -3$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length $\frac{1}{2}$. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] $\Delta t = \frac{1}{2n}$
- [ ] limits of integration
- [ ] integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{2n}$. Thus, the integral is over an interval of length $\frac{1}{2}$. This interval can be taken to be $\left[\frac{1}{4}, \frac{3}{4}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{4} + \frac{k}{2n}$ for k from 1 to n.

Therefore, $\lim_{n\to\infty}\left(\sum_{k=1}^{n} R\left(\frac{1}{4} + \frac{k}{2n}\right) \cdot \frac{1}{2n}\right) = \int_{\frac{1}{4}}^{\frac{3}{4}} R(t)\,dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+\frac{1}{2}} R\left(\frac{1}{4} - a + t\right)\,dt.$

6-3

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (weeks)	2	4	5	8	10
$L'(t)$ (feet per week)	1.0	0.8	0.7	0.4	0.2

The length of a vine during a 12-hour period is given by a twice-differentiable function L, where $L(t)$ is measured in feet and t is measured in weeks for $0 \le t \le 12$. The graph of L is concave down on the interval $0 \le t \le 12$. Selected values of the derivative of L, $L'(t)$, are given in the table above. At time $t = 4$, the length of the vine is 5 feet.

(a) Use the tangent line approximation for L at time $t = 4$ to estimate $L(4.3)$, the length of the vine at time $t = 4.3$. Is the approximation an overestimate or an underestimate for $L(4.3)$? Give a reason for your answer.

(b) Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_2^{10} L'(t)\,dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or an underestimate for $\int_2^{10} L'(t)\,dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^{n} L'\left(\frac{10k}{n}\right)\frac{10}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 4.3$. Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

6-3

The student response accurately includes both of the criteria below.

- [ ] approximation
- [ ] underestimate with reason

Solution:

$L(4.3) \approx L(4) + L'(4)(4.3 - 4) = 5 + 0.8(0.3) = 5.24$ feet

Since the graph of L is concave down on $4 \leq t \leq 4.3$, the tangent line lies above the graph of L on this interval. Therefore, the approximation is an overestimate for $L(4.3)$.

Part B

To earn the first point, the response must present the form of a left Riemann sum. To earn the second point, values for L' must be substituted into the sum. Therefore, $2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4$ earns the first and second points. Numerical answers do not need to be simplified for the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] left Riemann sum
- [ ] approximation
- [ ] units

Solution:

$$\int_2^{10} L'(t)\,dt \approx (4-2)L'(2) + (5-4)L'(4) + (8-5)L'(5) + (10-8)L'(8)$$
$$= 2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4 = 5.7 \text{ feet}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes overestimate with reason that references the behavior of L'

6-3

Solution:

Since the graph of L is concave down on $0 \leq t \leq 12$, L' is decreasing on $2 \leq t \leq 10$. Therefore, the left Riemann sum approximation is an overestimate for $\int_2^{10} L'(t)\, dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 10. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] $\Delta t = \frac{10}{n}$
- [ ] limits of integration
- [ ] integrand

Solution:

For this Riemann sum, $\Delta t = \frac{10}{n}$. Thus, the integral is over an interval of length 10. This interval can be taken to be $0 \leq t \leq 10$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{10k}{n}$ for k from 1 to n.

Therefore, $\lim_{n\to\infty}\left(\sum_{k=1}^{n} L'\left(\frac{10k}{n}\right)\frac{10}{n}\right) = \int_0^{10} L'(t)\, dt$.

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+10} L'(t-a)\, dt$.

6-3

12. Which of the following is equivalent to the definite integral $\int_2^6 \sqrt{x}\,dx$?

(A) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{4}{n}\sqrt{\frac{4k}{n}}$

(B) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{6}{n}\sqrt{\frac{6k}{n}}$

(C) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{4}{n}\sqrt{2+\frac{4k}{n}}$ ✓

(D) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{6}{n}\sqrt{2+\frac{6k}{n}}$

Answer C

$\Delta x = \frac{6-2}{n}$

13. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{3}{n}k\right)^2\left(\frac{3}{n}\right)$ is

(A) 1

(B) 3

(C) 9 ✓

(D) nonexistent

Answer C

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of width $\Delta x = \frac{3}{n}$ over an interval of width 3. The first right endpoint is $\frac{3}{n}$ and the last right endpoint is $\frac{3n}{n} = 3$. This means that the integration can be taken over the interval $[0, 3]$ with the height of each rectangle determined by the function $f(x) = x^2$ evaluated at the right endpoints $x_k = \frac{3}{n}k = k\,\Delta\,x$ for k from 1 to n. The limit of the Riemann sum will be

$$\lim_{n\to\infty}\sum_{k=1}^{n} f(x_k)\,\Delta\,x = \int_0^3 f(x)\,dx = \int_0^3 x^2\,dx = \frac{x^3}{3}\Big|_0^3 = 9.$$

6-3

14. Which of the following expressions is equal to $\lim_{n\to\infty} \frac{1}{n}\left(\frac{1}{2+\frac{1}{n}} + \frac{1}{2+\frac{2}{n}} + \frac{1}{2+\frac{3}{n}} + \cdots + \frac{1}{2+\frac{n}{n}}\right)$?

(A) $\int_1^2 \frac{1}{x}\, dx$

(B) $\int_0^1 \frac{1}{2+x}\, dx$

(C) $\int_0^2 \frac{1}{2+x}\, dx$

(D) $\int_2^3 \frac{1}{2+x}\, dx$

Answer B

Correct. The sum inside the limit can be interpreted as a right Riemann sum in the form $\sum_{k=1}^{n}\left(\frac{1}{2+\frac{k}{n}}\right)\frac{1}{n} = \sum_{k=1}^{n} f(k\Delta x)\Delta x$, where $f(x) = \frac{1}{2+x}$ and $\Delta x = \frac{1}{n}$. The value of Δx corresponds to an interval of length 1. Since $k\Delta x = \frac{1}{n}$ for the initial value $k = 1$, the interval starts at 0. Therefore, the limit of the Riemann sum is equal to the definite integral $\int_0^1 f(x)dx = \int_0^1 \frac{1}{2+x}dx$.

6-3

15. Which of the following definite integrals is equal to $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{12k}{n}\cos\left(1+\frac{4k}{n}\right)\frac{4}{n}$?

(A) $\int_1^5 12\cos x\, dx$

(B) $\int_1^5 3x\cos x\, dx$

(C) $\int_0^4 12\cos(1+x)\, dx$

(D) $\int_0^4 3x\cos(1+x)\, dx$

Answer D

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of length $\Delta x=\frac{4}{n}$ over an interval of length 4. The right endpoints that are used in the sum must agree with both the term $1+\frac{4k}{n}=1+k\Delta x$ inside the cosine function and the term $\frac{12k}{n}=3k\Delta x$ multiplying the cosine function. These endpoints could be of the form $x_k=k\Delta x$ for k from 1 to n on the interval $[0,4]$. The Riemann sum would then be $\sum_{k=1}^{n}3x_k\cos(1+x_k)\Delta x=\sum_{k=1}^{n}f(x_k)\Delta x$, where $f(x)=3x\cos(1+x)$. The limit of the Riemann sum would be written as $\int_0^4 3x\cos(1+x)\,dx$.

6-3

16. Which of the following is a right Riemann sum for $\int_3^8 \sqrt{1+x}\,dx$?

(A) $\sum_{k=1}^{n} \left(\sqrt{1+\frac{k}{n}} \cdot \frac{1}{n}\right)$

(B) $\sum_{k=1}^{n} \left(\sqrt{4+\frac{k}{n}} \cdot \frac{1}{n}\right)$

(C) $\sum_{k=1}^{n} \left(\sqrt{1+\frac{5k}{n}} \cdot \frac{5}{n}\right)$

(D) $\sum_{k=1}^{n} \left(\sqrt{4+\frac{5k}{n}} \cdot \frac{5}{n}\right)$

Answer D

Correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{8-3}{n} = \frac{5}{n}$. The height of each rectangle is determined by $f(x) = \sqrt{1+x}$ evaluated at the right endpoints of the n subintervals. These endpoints are the x-values at $x_k = 3 + k\Delta x$ for k from 1 to n, since the integration starts at 3. The Riemann sum can be written as

$\sum_{k=1}^{n} f(x_k)\,\Delta x = \sum_{k=1}^{n} \sqrt{1+x_k} \cdot \Delta x = \sum_{k=1}^{n} \sqrt{1+3+k\Delta x} \cdot \Delta x = \sum_{k=1}^{n} \sqrt{4+k\Delta x} \cdot \Delta x$, where $\Delta x = \frac{5}{n}$.

6-3

17. Which of the following is a left Riemann sum approximation of $\int_1^7 (4\ln x + 2)\,dx$ with n subintervals of equal length?

(A) $\sum_{k=1}^{n}\left(4\ln\left(1+\frac{k-1}{n}\right)+2\right)\frac{1}{n}$

(B) $\sum_{k=1}^{n}\left(4\ln\left(\frac{6k}{n}\right)+2\right)\frac{6}{n}$

(C) $\sum_{k=1}^{n}\left(4\ln\left(1+\frac{6(k-1)}{n}\right)+2\right)\frac{6}{n}$

(D) $\sum_{k=1}^{n}\left(4\ln\left(1+\frac{6k}{n}\right)+2\right)\frac{6}{n}$

Answer C

Correct. For this integral, a left Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{7-1}{n} = \frac{6}{n}$. The height of each rectangle is determined by the function $4\ln x + 2$ evaluated at the left endpoints of the n subintervals. These endpoints are the x-values at $1 + (k-1)\Delta x$, for k from 1 to n, since the integration starts at 1. Since the left endpoints are being used, the first endpoint is 1, hence the need to use $k-1$ in the summation starting with $k = 1$. The Riemann sum can be written as $\sum_{k=1}^{n} f(x_k)\Delta x = \sum_{k=1}^{n}(4\ln(x_k)+2)\Delta x = \sum_{k=1}^{n}(4\ln(1+(k-1)\Delta x)+2)\Delta x$ where $\Delta x = \frac{6}{n}$.

6-3

18. Which of the following definite integrals are equal to $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(-2+\frac{8k}{n}\right)^3\frac{8}{n}$?

I. $\displaystyle\int_{-2}^{6} x^3\,dx$

II. $\displaystyle\int_{0}^{8} (-2+x)^3\,dx$

III. $\displaystyle\int_{0}^{1} 8(-2+8x)^3\,dx$

(A) I only

(B) II only

(C) III only

(D) I, II, and III

Answer D

Correct. This expression can be interpreted as the limit of a right Riemann sum of a function f with n terms built from rectangles of width $\Delta x=\frac{8}{n}$ over an interval of width 8.

If $f(x)=x^3$ over the interval $[-2,6]$ with $x_k=-2+k\Delta x=-2+\frac{8k}{n}$, then the limit of the Riemann sum is the definite integral in I, as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(-2+\frac{8k}{n}\right)^3\frac{8}{n}=\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=\int_{-2}^{6}f(x)\,dx=\int_{-2}^{6}x^3\,dx$$

If $f(x)=(-2+x)^3$ over the interval $[0,8]$ with $x_k=k\Delta x=\frac{8k}{n}$, then the limit of the Riemann sum is the definite integral in II, as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(-2+\frac{8k}{n}\right)^3\frac{8}{n}=\lim_{n\to\infty}\sum_{k=1}^{n}(-2+x_k)^3\Delta x$$

$$=\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=\int_{0}^{8}f(x)\,dx=\int_{0}^{8}(-2+x)^3\,dx$$

The limit can also be interpreted as the limit of a right Riemann sum of a function f with n terms built from rectangles of width $\Delta x=\frac{1}{n}$ over an interval of width 1.

If $f(x)=8(-2+8x)^3$ over the interval $[0,1]$ with $x_k=k\Delta x=\frac{k}{n}$, then the limit of the Riemann sum is the definite integral in III, as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(-2+\frac{8k}{n}\right)^3\frac{8}{n}=\lim_{n\to\infty}\sum_{k=1}^{n}(-2+8x_k)^3 8\Delta x$$

$$=\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=\int_{0}^{1}f(x)\,dx=\int_{0}^{1}8(-2+8x)^3\,dx$$

6-3

19. Which of the following limits is equal to $\int_2^5 \sqrt{1+x^4}\,dx$?

(A) $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sqrt{1+\left(\frac{k}{n}\right)^4}\right)\frac{1}{n}$

(B) $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sqrt{1+\left(\frac{3k}{n}\right)^4}\right)\frac{3}{n}$

(C) $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sqrt{1+\left(2+\frac{k}{n}\right)^4}\right)\frac{1}{n}$

(D) $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sqrt{1+\left(2+\frac{3k}{n}\right)^4}\right)\frac{3}{n}$

Answer D

Correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{5-2}{n} = \frac{3}{n}$. The height of each rectangle is determined by the function $\sqrt{1+x^4}$ evaluated at the right endpoints of the n intervals. These endpoints are the x-values at $2+k\Delta x$ for k from 1 to n, since the integration starts at 2. The Riemann sum can be written as $\sum_{k=1}^{n}\sqrt{1+(2+k\Delta x)^4}\Delta x$ where $\Delta x = \frac{3}{n}$, and its limit as $n \to \infty$ is $\int_2^5 \sqrt{1+x^4}\,dx$.

6-3

20. $$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sin^2\left(1+\frac{4k}{n}\right)+3\right)\left(\frac{4}{n}\right)=$$

(A) $\int_0^4 \left(\sin^2 x+3\right) dx$

(B) $\int_1^5 \left(\sin^2 x+3\right) dx$

(C) $4\int_0^1 \left(\sin^2(1+x)+3\right) dx$

(D) $\int_0^1 \left(\sin^2(1+4x)+3\right) dx$

Answer B

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of width $\Delta x=\frac{4}{n}$ over an interval of length 4. The first (right) endpoint is when $k=1$, which corresponds to $1+\frac{4}{n}$. The last (right) endpoint is when $k=n$, which corresponds to $1+\frac{4n}{n}=1+4=5$. This means the integration can be taken over the interval $[1,5]$ with the height of each rectangle determined by the function $\sin^2 x+3$ evaluated at the right endpoints of the n intervals. The limit of the Riemann sum will be equal to $\int_1^5 \left(\sin^2 x+3\right) dx$.

6-3

21. Which of the following is a right Riemann sum for $\int_1^4 \arctan(1+x)\,dx$?

(A) $\sum_{k=1}^{n}\left(\arctan\left(1+\frac{k}{n}\right)\cdot\frac{1}{n}\right)$

(B) $\sum_{k=1}^{n}\left(\arctan\left(4+\frac{k}{n}\right)\cdot\frac{1}{n}\right)$

(C) $\sum_{k=1}^{n}\left(\arctan\left(1+\frac{3k}{n}\right)\cdot\frac{3}{n}\right)$

(D) $\sum_{k=1}^{n}\left(\arctan\left(2+\frac{3k}{n}\right)\cdot\frac{3}{n}\right)$

Answer D

Correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x=\frac{4-1}{n}=\frac{3}{n}$. The height of each rectangle is determined by $f(x)=\arctan(1+x)$, evaluated at the right endpoints of the n subintervals. These endpoints are the x-values at $x_k=1+k\Delta x$ for k from 1 to n, since the integration starts at 1. The Riemann sum can be written as

$$\sum_{k=1}^{n} f(x_k)\,\Delta x=\sum_{k=1}^{n}\arctan(1+x_k)\cdot\Delta x=\sum_{k=1}^{n}\arctan(1+1+k\Delta x)\cdot\Delta x=\sum_{k=1}^{n}\arctan(2+k\Delta x)\cdot\Delta x,$$

where $\Delta x=\frac{3}{n}$.

6-3

22. Which of the following is a left Riemann sum approximation of $\int_2^8 \cos\left(x^2\right) dx$ with n subintervals of equal length?

(A) $\sum_{k=1}^{n}\left(\cos\left(2+\frac{k-1}{n}\right)^2\right)\frac{1}{n}$

(B) $\sum_{k=1}^{n}\left(\cos\left(\frac{6k}{n}\right)^2\right)\frac{6}{n}$

(C) $\sum_{k=1}^{n}\left(\cos\left(2+\frac{6(k-1)}{n}\right)^2\right)\frac{6}{n}$

(D) $\sum_{k=1}^{n}\left(\cos\left(2+\frac{6k}{n}\right)^2\right)\frac{6}{n}$

Answer C

Correct. For this integral, a left Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{8-2}{n} = \frac{6}{n}$. The height of each rectangle is determined by the function $\cos\left(x^2\right)$ evaluated at the left endpoints of the n subintervals. These endpoints are the x-values at $2+(k-1)\Delta x$, for k from 1 to n, since the integration starts at 2. Since the left endpoints are being used, the first endpoint is 2, thus the need to use $k-1$ in the summation starting with $k=1$. The Riemann sum can be written as $\sum_{k=1}^{n} f(x_k)\Delta x = \sum_{k=1}^{n}\left(\cos(x_k)^2\right)\Delta x = \sum_{k=1}^{n}\left(\cos\left(2+(k-1)\Delta x\right)^2\right)\Delta x$ where $\Delta x = \frac{6}{n}$.

6-3

23. Which of the following definite integrals are equal to $\lim_{n\to\infty}\sum_{k=1}^{n}\sin\left(-1+\frac{5k}{n}\right)\frac{5}{n}$?

I. $\int_{-1}^{4}\sin x\,dx$

II. $\int_{0}^{5}\sin(-1+x)\,dx$

III. $5\int_{0}^{1}\sin(-1+5x)\,dx$

(A) I only

(B) II only

(C) III only

(D) I, II, and III

Answer D

Correct. All three expressions can be interpreted as limits of right Riemann sums of a function f with n terms built from rectangles of the appropriate width, Δx.

I. If $f(x)=\sin x$ over the interval $\left[-1,4\right]$, $\Delta x=\frac{4-(-1)}{n}=\frac{5}{n}$, and $x_k=-1+k\Delta x=-1+k\left(\frac{5}{n}\right)$, then the limit of the Riemann sum is the definite integral as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\sin\left(-1+\frac{5k}{n}\right)\frac{5}{n}=\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=\int_{-1}^{4}\sin x\,dx$$

II. If $f(x)=\sin(-1+x)$ over the interval $\left[0,5\right]$, $\Delta x=\frac{5-0}{n}=\frac{5}{n}$, and $x_k=-1+k\left(\frac{5}{n}\right)$, then the limit of the Riemann sum is the definite integral as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\sin\left(-1+\frac{5k}{n}\right)\frac{5}{n}=\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=\int_{0}^{5}\sin(-1+x)\,dx$$

III. If $f(x)=\sin(-1+5x)$ over the interval $\left[0,1\right]$, $\Delta x=\frac{1-0}{n}=\frac{1}{n}$, and $x_k=-1+5k\left(\frac{1}{n}\right)$, then the limit of the Riemann sum is 5 times the integral as follows.

$$\lim_{n\to\infty}\sum_{k=1}^{n}\sin\left(-1+\frac{5k}{n}\right)\frac{5}{n}=5\lim_{n\to\infty}\sum_{k=1}^{n}\sin\left(-1+\frac{5k}{n}\right)\left(\frac{1}{n}\right)=5\lim_{n\to\infty}\sum_{k=1}^{n}f(x_k)\Delta x=5\int_{0}^{1}\sin(-1+5x)\,dx$$

6-3

24. Which of the following definite integrals is equal to $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{10k}{n}\left(\sqrt{1+\frac{5k}{n}}\right)\left(\frac{5}{n}\right)$?

(A) $\int_1^6 10\sqrt{x}\,dx$

(B) $\int_1^6 2x\sqrt{x}\,dx$

(C) $\int_0^5 10\sqrt{1+x}\,dx$

(D) $\int_0^5 2x\sqrt{1+x}\,dx$

Answer D

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of length $\Delta x=\frac{5}{n}$ over an interval of length 5. The right endpoints that are used in the sum must agree with both the term $1+\frac{5k}{n}=1+k\Delta x$ inside the square root function and the term $\frac{10k}{n}=2k\Delta x$ multiplying the square root function. These endpoints could be of the form $x_k=k\Delta x$ for k from 1 to n on the interval $[0,5]$. The Riemann sum would then be $\sum_{k=1}^{n}2x_k\sqrt{1+x_k}\Delta x=\sum_{k=1}^{n}f(x_k)\Delta x$, where $f(x)=2x\sqrt{1+x}$. The limit of the Riemann sum would be written as $\int_0^5 2x\sqrt{1+x}\,dx$.

6-3

25. Which of the following limits is equal to $\int_{1}^{3} \sin\left(x^{3}+2\right) dx$?

(A) $\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\left(\frac{k}{n}\right)^{3}+2\right)\frac{1}{n}$

(B) $\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\left(\frac{2k}{n}\right)^{3}+2\right)\frac{2}{n}$

(C) $\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\left(1+\frac{k}{n}\right)^{3}+2\right)\frac{1}{n}$

(D) $\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\left(1+\frac{2k}{n}\right)^{3}+2\right)\frac{2}{n}$

Answer D

Correct. For this integral, a right Riemann sum with n terms is built from rectangles of width $\Delta x = \frac{3-1}{n} = \frac{2}{n}$. The height of each rectangle is determined by the function $\sin\left(x^{3}+2\right)$ evaluated at the right endpoints of the n intervals. These endpoints are the x-values at $1+k\Delta x$ for k from 1 to n, since the integration starts at 1. The Riemann sum can be written as $\sum_{k=1}^{n} \sin\left((1+k\Delta x)^{3}+2\right)\Delta x$ where $\Delta x = \frac{2}{n}$, and its limit as $n \to \infty$ is $\int_{1}^{3} \sin\left(x^{3}+2\right) dx$.

6-3

26. $$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt[4]{2+\frac{3k}{n}}\cdot\frac{3}{n}\right)=$$

(A) $\displaystyle\int_0^3 \sqrt[4]{x}\, dx$

(B) $\displaystyle\int_2^5 \sqrt[4]{x}\, dx$ ✓

(C) $\displaystyle 3\int_0^1 \sqrt[4]{2+x}\, dx$

(D) $\displaystyle\int_0^1 \sqrt[4]{2+3x}\, dx$

Answer B

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of width $\Delta x = \frac{3}{n}$ over an interval of length 3. The first (right) endpoint is when $k = 1$, which corresponds to $2 + \frac{3}{n}$. The last (right) endpoint is when $k = n$, which corresponds to $2 + \frac{3n}{n} = 5$. This means the integration can be taken over the interval $[2, 5]$ with the height of each rectangle determined by the function $\sqrt[4]{x}$ evaluated at the right endpoints of the n intervals. The limit of the Riemann sum will be equal to $\displaystyle\int_2^5 \sqrt[4]{x}\, dx$.

6-3

27. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \cos\left(\sqrt{x}\right)$.

(a) Approximate the definite integral $\int_1^3 f(x)\,dx$ using a midpoint Riemann sum with the subintervals $[1, 1.6]$, $[1.6, 2]$, and $[2, 3]$. Show the work that leads to your answer.

(b) Approximate the definite integral $\int_1^3 f(x)\,dx$ using a trapezoidal sum with the subintervals $[1, 1.6]$, $[1.6, 2]$, and $[2, 3]$. Show the work that leads to your answer.

(c) It is known that $f'(x) = -\frac{\sin(\sqrt{x})}{2\sqrt{x}}$ and $f''(x) = \frac{\sin(\sqrt{x})}{4x\sqrt{x}} - \frac{\cos(\sqrt{x})}{4x}$. Would a trapezoidal sum approximation for $\int_1^3 f(x)\,dx$ overestimate or underestimate the value of $\int_1^3 f(x)\,dx$? Give a reason for your answer.

(d) Write $\int_1^3 f(x)\,dx$ as the limit of a right Riemann sum with n subintervals of equal length.

Part A

The first point is earned for correctly presenting the form of the midpoint sum.

The second point is earned for correctly substituting the x -values into the expression for f in the sum. Therefore, $(0.6) \cdot \cos\left(\sqrt{1.3}\right) + (0.4) \cdot \cos\left(\sqrt{1.8}\right) + (1) \cdot \cos\left(\sqrt{2.5}\right)$ earns both points. Numerical answers must be accurate to three places after the decimal point and may be rounded or truncated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-3

0	1	2

The student response accurately includes both of the criteria below.

- [ ] midpoint sum
- [ ] approximation

Solution:

$$\int_1^3 f(x)\,dx \approx (1.6-1)\cdot f(1.3) + (2-1.6)\cdot f(1.8) + (3-2)\cdot f(2.5)$$

$$= (0.6)\cdot \cos\left(\sqrt{1.3}\right) + (0.4)\cdot \cos\left(\sqrt{1.8}\right) + (1)\cdot \cos\left(\sqrt{2.5}\right)$$

$$= 0.331 \text{ (or } 0.330)$$

Part B

The first point is earned for correctly presenting the form of the trapezoidal approximation. The second point is earned for correctly substituting the x -values into the expression for f in the sum. Therefore,

$$0.3\cdot\left(\cos\left(\sqrt{1}\right)+\cos\left(\sqrt{1.6}\right)\right)+0.2\cdot\left(\cos\left(\sqrt{1.6}\right)+\cos\left(\sqrt{2}\right)\right)+0.5\cdot\left(\cos\left(\sqrt{2}\right)+\cos\left(\sqrt{3}\right)\right)$$

earns both points. Numerical answers must be accurate to three places after the decimal point and may be rounded or truncated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] trapezoidal approximation
- [ ] approximation

Solution:

6-3

$$\int_1^3 f(x)\,dx \approx (1.6-1)\cdot\frac{f(1)+f(1.6)}{2} + (2-1.6)\cdot\frac{f(1.6)+f(2)}{2} + (3-2)\cdot\frac{f(2)+f(3)}{2}$$

$$= 0.3\cdot\left(\cos\left(\sqrt{1}\right)+\cos\left(\sqrt{1.6}\right)\right) + 0.2\cdot\left(\cos\left(\sqrt{1.6}\right)+\cos\left(\sqrt{2}\right)\right) + 0.5\cdot\left(\cos\left(\sqrt{2}\right)+\cos\left(\sqrt{3}\right)\right)$$

$$= 0.342 \text{ (or } 0.341\text{)}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ considers $f''(x)$
- ☐ overestimate with reason

Solution:

$f''(x) > 0$ for $1 < x < 3$.

$\Rightarrow$ The graph of f is concave up for $1 < x < 3$.

A trapezoidal sum is an overestimate because the graph of f is concave up for $1 < x < 3$.

(Note: Since the increasing/decreasing behavior of a function does not affect whether a trapezoidal sum is an overestimate or an underestimate, the first derivative f' is not taken into consideration.)

Part D

The first point may be earned as presented in the solution or by using that information to determine that the subintervals are of length $\frac{2}{n}$. A response that produces the correct expression earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

6-3

- ☐ $\Delta x = \frac{2}{n}$
- ☐ summation notation with limit as $n \to \infty$
- ☐ summand

Solution:

$\Delta x = \frac{3-1}{n} = \frac{2}{n}$

$$\int_1^3 f(x)\,dx = \lim_{n\to\infty}\left(\sum_{k=1}^{n} f(1+k\Delta x)\cdot\Delta x\right) = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\cos\sqrt{1+\frac{2k}{n}}\cdot\frac{2}{n}\right)$$

Note: The definite integral can be written as any limit of a Riemann sum of the form

$$\lim_{n\to\infty}\left(\sum_{k=a}^{n+a-1}\cos\sqrt{1+\frac{2}{n}(k-a+1)}\cdot\frac{2}{n}\right).$$